SI411: Why the Best Portfolios Are Built to Be Wrong ft. David Dredge & Richard Brennan

Podcast: Top Traders Unplugged

AI Executive Summary

This transcript features a deep-dive conversation between **Niels (the host)**, **Dave (a long-volatility/convexity specialist)**, and **Rich (a trend follower)**. The discussion centers on rethinking traditional risk management through the lens of "convexity," "fractal market structures," and "asymmetric returns."

Below is a thematic summary of the key arguments presented:

1. Redefining Diversification

The speakers argue that traditional diversification can be dangerous because it often relies on false assumptions about correlation. * **The Danger of "Single Anchor" Diversification:** Rich uses the analogy of thousands of strands crossing a canyon. If all those strands are connected to the same two anchors (e.g., all assets tied to interest rate moves), you aren't diversified; you have one single failure mode. * **Permanent Coupling:** Contrary to the belief that markets only correlate during crises, the speakers argue that markets are **permanently coupled** through the very act of investors diversifying across products. During "calm" periods, this coupling manifests as oscillation (diversification appears to work); during "crisis" periods, it shifts into a trend (everything moves together). * **Diversification as an Engine for the "Right Tail":** Rather than using diversification to *mitigate risk* (which can lead to "thinning" only the middle of the distribution), they use it to **expand exposure to rare, massive opportunities.** Rich notes that by being "maximally diversified," they held cocoa when it explodedâ€”an idiosyncratic opportunity a smaller, more concentrated portfolio would have missed.

2. The Concept of Convexity and Asymmetry

The core investment philosophy shared by Dave and Rich is the pursuit of **positive convexity** (the "accelerator"). * **Truncating the Left Tail:** Both strategies aim to limit downside ("the brakes"). Dave does this through

explicit option premiums (explicitly paying for protection), while Rich does it via stop-losses (implicitly paying a "premium" through whipsaws and lost opportunities). able **Capturing the Right Tail:** By protecting the downside, they can leave the upside unbounded. They seek to participate in "infinite" potential gains (the St. Petersburg Paradox approach) where one massive trade can fund the entire budget forever. * **The "Triple Whammy":** Dave explains how he seeks mispriced volatility by buying long-dated, non-linear products, benefiting from both price mispricing and the flaws in how volatility is supplied to the market.

3. Practical Advice for Investors (Monday Morning Actions)

The speakers offer specific, contrarian advice for fund managers and individual investors:

From Dave (Focus on Structure and Incentives): * **Change the Incentive Structure:** Move away from arithmetic/calendar-year returns toward **geometric compounding**. * **Add Convexity:** Don't just optimize; add "brakes" to your portfolio so you can safely drive faster. * **Free up Capital:** Stop using capital as "ballast" to slow yourself down. Use "brakes" (protection) to free that capital for more aggressive, upside-participating assets.

From Rich (Focus on Execution and Scaling): * **Cap Leverage in Absolute Terms:** Break the link between a "quiet market" and "large position sizes." High leverage in quiet markets is the primary driver of catastrophic wealth destruction. * **Size by Risk, Not Strength:** Don't tilt toward whatever looks strongest recently (which introduces selection bias). Size every market to the same risk level to maintain true breadth. * **Grow by Width, Not Weight:** When you want more return, don't increase the size of existing bets (concentration); add new, unrelated markets/systems (diversification). * **Enter Small:** Let the market build a position for you using "open profit" rather than trying to force your will on the market with large initial entries.

4. The Role of Accountability

The conversation concludes with a critique of modern risk management. The speakers argue that **true risk is "everything you didn't think of."** They contend that much of modern financial regulation (like VaR-based capital requirements) fails because it removes **accountability** from decision-makers, replacing human judgment and skin-in-the-game with flawed, backward-looking statistical models.

Full Transcript

Welcome to Top Traders Unplugged. In markets, success doesn't come from predicting what happens next. It comes from being prepared for what you can't predict. In each episode, we go deep with some of the world's most thoughtful minds in investing, economics, and beyond to understand how they think, how they prepare, and how they decide, and the experiences that shaped how they see the world. No noise, no shortcuts, just real conversations to help you think better and invest with confidence. Welcome or welcome back to this week's edition of the Systematic Investor Series with Richard Brennan and I, Niels Kastur Larsen, where each week we take the pulse of the global markets through the lens of a rules-based investor. This week we have a very special guest, namely the one and only Dave Dredge, who really needs no introduction but who officially is the Chief Investment Officer at Convex Strategies in Singapore.

Rich and Dave, it is really wonderful to have you both here this week. Hope you're doing well. What's going on in your part of the world, so to speak? I'll let Dave go first since he's the... It's summer in Singapore, which is just like any other time of the year in Singapore. So it's hot and humid and rains a lot. But business-wise, we're super busy. There's obviously a lot going on in the world. And then the need for... effective explicit loss mitigation, effective explicit negatively correlating explicit true diversifying has never been higher. And so we're pretty busy and markets are active across all kinds of asset classes. And I've got some presentations coming up. I'm down in Australia in a couple of weeks, speaking at a conference down there. So, everything's going here as usual. That's fantastic, Gustav. How about you, Rich? How are you doing?

It's a bit colder for me, Niels. So down in the southern hemisphere here, I'm not too far away from Dave in Singapore, but sufficiently south to know it's winter down here. So I noticed that Dave and I both got our merch on for top traders unplugged. And the only one who's absent of that Niels is you. And I assume it's because it's the boiling summer heat up there. And you can't afford to have anything over that top of yours. Well, I mean, that is definitely one excuse. The other more truthful excuse is I completely forgot

it. But there we are. I will have to improve my prep skills. That is for sure. It's great to have you here. And it is really wonderful knowing what topics we're going to be talking about, which I appreciate both of you help put together. And I think really the audience is in for a treat this week. Now...

It is coming up to month end in July and my trend barometer yesterday finished at 45, which is pretty neutral, probably more or less in line with what we see in terms of the indices. And for those who listen on a regular basis, you'll notice that I skipped the what's been on your radar because we got so much stuff to cover with Dave and Rich that I thought we would just simply go on to that a little bit quicker today. Now, we're recording on Thursday. We've got one and a half days left of the month of July. So we'll see how it goes. But so far, the data looks like the following. And this is as of Tuesday evening, BTOP 50 down 23 basis points up 8.32% for the month. Sorry, for the year. SGCTA index down 79 basis points for the month and up 8.5% for the year. SOCGEN trend down 75 basis points for the month up 8.3% for the year.

And the short-term traders index that's struggling a bit more down 1.71% in July and up 3.34% so far this year. In the traditional world, MSCI world is down 1.63%. And if we exclude US and Canada, which was suggested by Andrew, not the Andrew that comes on as a co-host, but another Andrew, that is down 26 basis points only for the month and up 9.56% so far this year. The S&P US aggregate bond index down 91 basis points. So no relief on that. And it's flat for the year. And the S&P 500 total return down 2.39% so far in July, up 7.58% so far this year. Before we dive into all the topics, just wanted to ask whether there was anything from the month of July. I mean, obviously there's been some activity, I think, to say the least over in Taiwan, if I'm not mistaken. But anything else going on from your vantage point? Well, there's a lot going on in my world. So you've got sort of multi-year, multi-decade highs in long-dated bond yields in Japan, in the US, in Britain. You've got a whole bunch of central bank policy meetings over the end of the month, obviously the Fed overnight last night.

standing pat with three descents in the vote, very unusual there, and a huge disappointment in the bond market. You've had a ton of noise out here, as you mentioned, across some of the equity markets. Yeah, I said Taiwan,

Dave, but I meant Korea, didn't I? Yeah, Korea, but Taiwan also. Taiwan hasn't been quiet. Japan hasn't been quiet. And so, in particular, the big semi-chip manufacturers in Japan, in Taiwan, in Korea. But Korea, most particularly, where you've had this proliferation of levered ETFs that has fueled this boom bus there that's been extraordinary. I was just looking at it. These two times levered ETFs in Hong Kong on the listed chip makers in Korea. So one of those being SK Hynix. So just over the last three months, SK Hynix itself as of the close today is up 1%. The two times levered ETF is down 47%.

Wow. And so it's such a perfect example, and I'm sure we'll talk about it, of what in the business is known as volatility drag. And so at some point in that three month period, the underlying stock was up as much as 125% and is now all the way back down to up 1%. But the volatility drag of the 2X leverage, which is a 2X the daily move, not 2X over the period, You know has been up as much as 300% it is now down 47% from the start and As something I say all the time and rich will get this particularly Over a long enough time horizon all levered ETFs approach zero now and not to not to to hijack the conversation on that topic right now, but You kind of wonder why authorities keep giving people these nuclear instruments, right? I mean, you really have to wonder. In fact, Korea is talking a lot today. There's an article today from one of the big normal ETF managers in Korea saying, we shouldn't allow these things. These things are, again, they're just volatility drag. It's just transferring wealth away from the end investor.

it's inevitable that way. It's simply the mathematics of geometric returns and the impact of that variance on the drawdown versus the drawup over time. It's inevitable that they go to zero. Anything in your part of the world, Rich, that we want to highlight before we move on? It's just been a volatile time for us, Neil. So it's a bit like... What is it? Those magnificent men in their flying machines goes up and then it goes down and then it goes up and then turns around. So that's what we're getting for this month, especially with the Cosby. That was a significant decline. So it's been a challenging couple of months for us, Niels. Yeah, absolutely. All right, let's move on. Let me try and set up the conversation the best I can. And then we move into these questions, topics that we prepared.

Obviously, Rich, you've been a regular contributor to the podcast for many years, and Dave has been one of the most popular guests each time he has joined the show, but something has been kind of nagging me over the years. Now, you both come at markets from completely different directions. So Dave from the world of options, convexity and tail protection, and Rich, you come from trend following and quantitative research. Different backgrounds. different tools, different terminology entirely actually. And yet when I think back to all those conversations we've had, the two of you sound like echoes of each other. So I went looking for why and I think I found it. Both of you see markets as complex adaptive systems rather than as statistical distributions to be sampled and estimated. That one commitment and almost everything else you believe seems to follow from that.

So that is what I want to explore today. Take that idea, examine it properly and follow it wherever it goes because it takes both of you to the same uncomfortable conclusion that the wrong mathematics is being deployed in markets and that risk is not volatility. It's something deeper than that. So I'm going to put questions to each of you knowing full well that we will most likely end up going a little bit different directions, and it'll be more of a fluent conversation. But anyways, I hope you're up for it. And this may turn out to be kind of the lecture where the student being me have two professors to ask questions to. And we'll feel a little bit underprepared as the lecture begins. But there we are. Wonderful. Anyways, let's start with kind of the shared lineage. So Dave.

What do you actually mean when you say that market is a complex adaptive system? And what does the conventional view assume it to be? I think I'll give rich his flowers I think he said it best maybe ever on one of your interviews with him when he said the the future isn't hidden It hasn't been written yet. Hmm. True. We don't know what it is because there's too many moving parts And so the way we think of it, and we talk about it here, is Ed Lorenza's butterfly wings, that it's infinite paths inside a finite space chasing an unattainable stranger tractor. And our job is always to do everything we can to understand initial conditions and where we are. Understand the now and then be resilient to an unknown.

unknowable future path. And so you guys have heard before, everybody's heard my race car analogy. It's not about forecasting the future circumstances of the race. It's about making your car as resilient as it can possibly be. And then being able to react and respond to the unknowable path of the future. Because the future is so much more than simply the shape of the race course. It's what other drivers doing. It's what you're doing. whether it's wetness, it's wear and tear, it's all these things that are impossible to predict that all interact and affect each other. And the, as Rich will say so eloquently, the emergent outcome of all those different parts that create a whole that is unexplainable by the individual parts. And so we see the world that way, and then we see our task on behalf of the investors that we work with is to prepare them with a car that has breaking capacity, that makes it the best car for achieving that resilience and the ability to accelerate and decelerate and react as things occur in the ever-evolving initial conditions, chasing that strange attractor through this infinite loop of a finite universe.

No, absolutely. Now, as Dave alluded to, Rich, you were right at the same place, but in a completely different way. What has convinced you about this? or first big fan of yours Dave have been a long-term big fan of yours and sort of I'll quietly go to sleep at night reading one of your memos and yeah I just feel as though we're very much aligned in the same space but my interest stems from a strong passion in science which co-evolved with my strong interest in finance but recognising that the two weren't aligned. I think you see that science has progressed from the world of linear outcomes, the world of Newton into a world that's much more dynamic, a lot less stationary. It's a world of reflexivity, where matter tells space how to curve and space tells matter how to move. There's this combined reflexivity, which Einstein found, which overturned Newton.

And that therefore changed concepts such as what is length because length became a very frame dependent posture. And I feel that finance is sort of maybe 100 years behind where the science is today in their understanding of what these systems are. When we talk about complex adaptive systems, When we use the word complex and when we use the word adaptive, I don't think people really understand what we're talking about there. We are truly talking about complexity and non-stationarity and evolution of systems.

We're talking about processes here. We're not talking about a world of things. We're talking about a world of processes. And these processes are very contingent on constraints. And when you map, all of the things that can affect an outcome, which is typically what we see in the statistical residue of the outcomes. That's what I call a statistical residue of a process. The deep process itself shows that there are so many multivariate variables affecting the outcome that it is impossible to forecast an outcome with a degree of fidelity.

So the way I see these systems, finance has gone back to the world of Euclid, linearity, the straight line. I think Mandelbrot first exposed that in 1960, showing that there was roughness at every scale in the financial markets, a bit like the coastlines of England. And you couldn't use a standard ruler to measure at each scale because each gave different results. We had to look at the dimension or the scaling property that existed between those lengths to then understand how that system sort of could be understood in terms of its volatility and those sort of things. But finance has really resisted that move for such a long time. It's stuck in this world of Euclid.

where it's a maths, it's a correct form of maths, but it's not a maths to apply to this domain we call the reality, which is the financial markets we observe today. two key things. Firstly, there's roughness at every scale. There's a fractal geometry. There's these attractors, as Dave so eloquently put, we have a system that is computationally irreducible. So what that means is that, as Dave mentioned, the complexity and the decisions that, especially in a reflexive system where you get the agents within the system reacting and adapting to the system and the system forcing that adaption and reaction. There's this reflexivity involved which makes our ability to deal with simple linear rulers non-functional. We'll find that the system that we are inquiring and investigating is inherently non-linear. It's unpredictable. The future is being written at every step and to understand what happens in five days time, you need to understand the decisions and consequences of all of the agents who'll be transacting over that period and their conglomerated positions and all of their effects, which is incalculable from the domain.

sits in the now and therefore, it's not like a future that is hidden in fog. It's a future that is inherently unpredictable. So I view these systems, particularly these what we call complex adaptive systems, which are particularly inherent in living systems, systems where we have agents who react to that system and the system reacts to those agents as this dynamic reflexivity. When we have that occurring, which is typically in living systems, we find that they are complex adaptive systems. It's a bit different to say the weather, which is a physical system. The molecules in the weather, they're not reflexive in the same sense that an agent in our market is. And I'll go back to that famous quote by Richard Feynman. He said, imagine how hard physics would be if electrons have feelings. Now that, is what the market is. And that's why it is a complex system and incredibly difficult to crack. And despite the Nobel laureates we've had, the very clever quants, you will be really surprised at the fact that there is very little evidence that anyone has significantly beaten the market. The market is the master. And, you know, I remember that movie Margin Call with Kevin Spacey and Jeremy Irons, Jeremy Irons was eating his steak recounting the litany of collapses that have occurred over the last 200 years. And how, you know, these very clever people have always sort of experienced these collapses without being able to resolve why these occurred. And it's because they're using a mathematics that is not appropriate for the domain and reality we exist in. So that's where my interest is. That's why I think Dave and I are very aligned in our views. Yeah. Well, excellent. Great stuff.

All right. So obviously, we're in the world of investing that is directional. We like when price moves. So let me sort of focus on that a little bit. We know, and we know that also from the work of people like Mike Green, that a lot of money has been flowing into the markets recently. That never looks really at the price, but it has an impact on price, I guess. So I wonder maybe from your perspective, Dave, what What does that do to price that you have this, and I call it new. It's not that new, but it's been there for a little while. This type of investor that looks at price differently, meaning price insensitive. Yeah. This is something we could go on about for the entire time period. As I think you know, I'm reasonably good friends with Mike Green and with Ari Krishnan and huge fan of their work.

and their most recent paper on passive investing and equity flows, which they were kind enough to even give me an acknowledgement in for some feedback I had provided. I'm known for years and years of what some people fondly refer to as dredgisms in the world. One of my famous dredgisms is positioning is the only thing that matters. I love all the time where people want to justify why a market moved or this or that reason, positioning is the only thing that matters. And sort of that passive flow dynamic. And again, something that I know you and Rich talk about all the time, talking about price and sensitive investors versus price sensitive investors. Similarly, when we model markets, we break markets into, and again, this will sound very familiar to you guys.

we break them into what I call rational accounting man, which would be price insensitive, people that are following accounting rules and regulation or instructions that are non-economic related and what we would call boundedly rational agents, people who are making decisions based upon the risk they're taking, skin in the game, their own capital, tacit knowledge. model things that way. And so you can think of that passive as a subset of bounded, of sorry, rational accounting man. And rational accounting man dominates most markets because all bank traders, all pension fund traders, all insurance traders, all central bank traders, all FX reserve managers are rational accounting men. They're not, when they buy something they're not thinking about the value of it for their retirement. They're thinking about the regulatory requirements around accounting and risk and a usually an annual compensation period. That is a very different thought process than somebody who's trying to make a 30-year investment decision to fund his retirement where losses are his and failures to achieve his investment goals impair his own lifestyle.

Those are two very different decision-making and very much what Rich talks about all the time. Now, one of the things I stress a lot to Mike and Hari on every time we talk about their work is, yes, I agree, but you're missing the biggest elephant in the room, which is the bond market. They talk about the percent of passive taking over and eventually becoming a problem when it gets beyond a certain amount, above and beyond. what we would call boundedly rational, which might call active agents. Well, the bond market went well through that at the peak of what I call the mother of

all bubbles of bond markets at the late 2020. And we saw the implications of that. And I'm sure, and I know you guys benefited on behalf of your investors, the implications of at least the initial shock unwind of that imbalance in 2022.

when interest rates went up and you had one of the biggest opportunities and trend that you've had in a long time. And I'll argue we're still in a big problem of unwinding that complexity because those rational accounting men, even in losses, do not behave like price sensitive investors because they don't mark the market. They have massive, the central banks who bought all the bonds are sitting on massive, unrealized losses. The FX reserve managers are sitting on massive unrealized losses. The banks are sitting on massive unrealized, says the pension funds and insurance companies that are a big part of some of the other side of stuff we do with duration, extending, callable note structures with embedded Bermudan's options that are masking duration in the form of structured complex derivative products have really massive unrealized losses and nobody's cut the positions yet.

Nobody's freed the capital that's tied up in those things or the lack of capital. Nobody's recognized the accounting losses. And so to me, this is one of the biggest drivers of all things in the financial system is exactly this. It's exactly what you and Rich talk about all the time is that interaction and the growing dominance at times of the price insensitive investor and of course the application of leverage in particular in the regulated space where Volatility is the measure of risk and low volatility means low risk and adding leverage doesn't increase the risk. The risk is the volatility no matter how much leverage you apply to the super senior structure of a sub-prem CDO. Now obviously Rich and I know very well that the leverage is the risk. It's the reason why it's a left skewed return distribution. It's the reason why tails are far fatter than a Gaussian normal distribution would.

hint at. The lack of capital is the stranger tractor at times pulling the market that direction because of the proliferation of loss management as that leverage has to get cleansed from the system. So to me, that's what moves prices. It's the accumulating imbalances as some sort of price and sensitive flow builds and builds on itself, reflexively building as the application of

leverage through a flawed regulatory construct, not just regulatory, but flawed understanding builds that and then reverses the other way with the fat tail on the other side as the mistake becomes recognized, the king has no clothes. The knowledge game recognizes that problem and it moves back the other way in what will be a very fat tail.

And that's the way we see the world. Yeah. No, no, absolutely. Now, you did an interesting exercise, Rich, because you built, quote unquote, a market where nobody really is responding to price. What happened with that? Well, I suppose this model was influenced by Jean-Philippe Bouchaud's sort of examination of micro structure. What moves price? And what I'd say is that, well, it's the transaction, the transaction event that moves price. And all other theories, you know, you might have fundamentals, you might have macro investing, you might have liquidity flows, all of these different theories, they're what are regarded as downstream of the transaction. The thing, the mechanistic thing that is actually moving price is the transaction itself. So the way I see these markets is a composite, it's an ecosystem of participants in those markets. And each of those particular And species in that ecosystem, they...

create a directed force based on the models that they adopt in that market. So fundamental investors will have a particular force implied to evaluation. Trend followers will be implying a sort of an amplifying force as they respond to price moving. So this therefore sort of comes to this point of what I call its feedback in the systems that creates this grapple structure and it's actually what is creating this movement in price. So when I look at the feedback, I see there's two forms of feedback. There's the positive feedback or what I call amplifying feedback and there is negative feedback, which is more the convergent premise. The reduction or the return to the mean or the mean reverting the countertrend.

fade. So these two forces, divergent and convergent, can be expressed by all of these different ecosystems of different species in these ecosystems and they all have these different forces being applied in the markets. Some are amplifying, some are destructive or mean reverting. Now when I put an agent-based model together, looking at what I call, let's look at the species of a market. I'm going to call one the random trader. They are the trader

who has no model that has a preference, a directional preference. They can be amplifying or they could be negative feedback. They're what I call the random trader. There's a convergent trader who's always fading the move. So markets are oscillating. They're fading the move at each way, trying to bring things back to what they consider an equilibrium-based position. Then there is the price sensitive. They're price sensitive. And then there's another price sensitive agent who I call the divergent trader. They're the ones that amplify the move.

Passive actually fits in that category. It's not that they are responding to price. They are price insensitive. They're responding to a rule, but that rule is invest in the biggest things. So what that does is it puts an amplifying pressure or directional force into that market. So with this agent-based model, what I did was I looked at just applying random participants in that model first and what I got was a Gaussian outcome. no fat tails, no memory, no volatility clustering. It was exactly what you'd, the outcomes you'd expect from the efficient market hypothesis, the Gaussian premise. But as soon as I started putting divergent participants in amplifying the move, I got to a position of about a quarter of the, about 25% and above of participants that are divergent.

All of the signatures you see in the real market automatically come back into the market. You get fat tails, you get volatility clustering, you get the memory, market memory. All of these signatures come back in once the agent-based model integrates a small percentage, only about 25% of these divergent participants. And passive, of course, is a divergent participant as well. So what I'm saying is that over the last 40 years, this has been a case. and this is why markets exhibit these properties, fat tails, validated clusters that maybe the orthodoxy has sort of refused to believe persists and they're regarding them as anomalies as opposed to a fundamental feature of market structure. So that's what I found in my modelling and it was very interesting that it wasn't a gradual sort of bring back of the signatures of the real market. It was a transition event. Once we achieved this 25% influence of divergent participants in there, it was a threshold and we got the typical phase transition as opposed to a gradual change. Before we get to that 25%, I can populate this simulated agent-based model with convergent traders, Gaussian. Random traders, Gaussian. Random and convergent, Gaussian.

As soon as I start putting those divergent population in there of 25% and above, boom, all the signatures come back into the market. So I found that a very fascinating bit of research that just tells me that over the history of these markets, this is a reality. And it's just this amplifying position. That's what's creating the tailors. That's what's creating the memory. That's what creates the volatility clusters. Yeah, another reality of market is that of course a lot of the time it's All moves noise and then from time to time we get the really big moves and some curious from your perspective Dave where does these Actually come from these sudden bigger bigger moves. How do they suddenly occur? it's Broadly speaking it's leverage but more simply speaking it's it's loss absorbing capital and so people because they're following very simplified rules of Gaussian linearity, ergodic single slices of time, they are taking more risk than they have the loss appetite for or the capital to absorb losses. And so when that mistake gets exposed because one of Rich's divergent traders comes in or something happens moving it, it triggers stop losses for the guys who miscalculated the risk. Now, I would say the biggest one I'll argue maybe in the whole system is the assumptions around stable correlation in what amounts to diversification. And so the world, the investment universe says we're happy with our risk appetite to take this much risk on an assumption of the diversification benefit.

created across these different assets, strategies, geographies, et cetera. And then when that correlation for whatever reason misbehaves, they've got more risk than they want. And so now they need to reduce risk. Now, of course, in that reflexive world that Rich speaks so well of, their reduction of risk in response to inconsistent correlation affects the correlation. which affects the risk appetite of the next guy, who then also has to adjust his risk, which then triggers the divergent trader who is saying, wait a minute, correlation's moving. I want to go with that moving correlation, which then affects the next guy. And so, again, the way we very much look at the world, the residual risk in most people's portfolios, I see all the time, most people do a pretty good job of managing the risk they manage.

or managing the risk they measure. But what they don't do a very good job of is managing the risk they assume, the risk they don't measure. And generally speaking, in large investment portfolios, that's correlation. And it's where they've made simplifying assumptions around correlation and

volatility, which are really the two things that everyone assumes constant in their simplified modern portfolio theory efficient market hypothesis methodology. is the thing that catches them out and now they have more risk than the capital they had allocated to absorb the losses. That of course gets more and more extreme as different types, either explicit leverage in the form of banks, balance sheets or hedge funds get utilized or the implicit leverage, obviously the world that we most active in of volatility selling, which creates then mispriced tails and the need to respond to things that you hadn't previously anticipated, which then leads to above normal standard deviation, what you would assume normal standard deviation market size moves. That's simply how we see the world. Yeah. Rich, these Indochinous forces, can they be measured in some way? Yeah.

Dave's forest fire analogy. The big moves that occur in the market, I believe, have an endogenous source as opposed to an exogenous source. It's not coming from news. News isn't actually creating the significant material moves. It's the state of the market. And the news might be the trigger, but it's the state of the market. Whether there's the undergrowth that's been building as Dave loves to say of his forest. forest fire analogy, totally understand that. So when you look at research like Jean-Philippe Bouchard, you see when he actually assessed whether it was news that was driving market microstructure. He came out explicitly saying that it was 90% of the moves were from endogenous sources. And what he was doing there at the, I think the tick by tick or the minute by minute, he was looking at the correlation between news events and market price movements from all sources of news events and seeing if there was a strong correlation between the two and he couldn't find it. And this is what I also observe in markets. It's the state of the market.

When it comes from exogenous or endogenous sources, it's not coming from nowhere. It's coming from within the market itself. And you find that it doesn't arrive out of nowhere. It builds progressively. So activity picks up beforehand. Liquidity quietly thins out. The book therefore gets very fragile. And then bang, it goes with a trigger. often takes also longer to settle afterwards. It's not instantaneous. So the traditional orthodoxy was that news events were instantaneously accepted by the market prices would adjust. But that's not the way there's a lag. And this lag, I think Gaby and

Cohen, a lot of research has been done on this that says that markets are very inelastic in their properties. And I think that it's this memory that exists in the market.

you know, markets have memory. And you find this when you examine what this volatility clustering is because when you do the research, you find that your calm begets calm and volatility begets volatility. They're amplifying the state of each other. Now, what happens during these calm periods, you get a progressive increase in the risk building up in the structure. And it's a transition between the calm state and the volatile state that get catches everyone naked and we'll go into that in later questions. But that's what I found in my models. I found that with no news events, I didn't need any news events in those agent-based models I was talking about. All I needed was the divergent participant, the convergent participant, the random participant. That's what was creating all of those signatures. It wasn't news that was doing that in my agent-based models. It was just the the impact that they're exerting with their models. That's what I found. Now, going into a little bit of deep water on my side, because both of you from time to time bring up this word, ergodicity, and I'm always kind of, what exactly is it we're talking about now? But anyway, from memory, I think it has something to do with ensemble averages versus time averages, something along those lines. Anyway, Dave, why is this important for people who have money at risk in the markets? I think it's important for a lot of things and certainly important for money at risk. And you're right, something that is ergodic, basically time doesn't matter. Sequencing doesn't matter, right? The ensemble average equals the time average. From a portfolio perspective, it would be the arithmetic mean equals the geometric mean.

but in something that is non-ergodic, that is not the case, right? And it turns out that the path through time makes an enormous difference. And so the easy one for everyone to think of is the usual problem that a 50% drawdown isn't corrected with a 50% recovery. You've got to get 100% recovery to get back. It's our, again, our race car example where it's a multi-lap race. And if you crash, you don't finish. It matters, right? Now, Nasim always tells the story about, you know, people go into the casino and if you give 30 people money to go play one casino game and they all come back,

well, they're gonna get the average. And if the 19th guy loses his money, it doesn't affect what happened to the 20th guy. But if you send one guy to go play 30 games and he loses all his money on the 19th game, he's out.

He doesn't get to play the 20th game. And so that's the nature of the investment process because it's a path through time. Sequencing matters is, again, another great Nasim quote. The average of the condition of your trousers is the same whether you launder them first or iron them first, but the end outcome is very, very different which way around you do it. Sequencing matters. It's a non-ergotic thing. And I, you know, I say it all the time and I'm sure you guys have either seen me write or say it. I think Ollie Peters, who's sort of the most prominent voice in the world of Ergodesi right now in his book, Ergodesi Economics, should be mandatory reading at every school level in mathematics, in physics, in finance, in economics, in business, in everything. It is the most beautiful mathematical description of common sense, of bounded rationality, of tacit knowledge that you will ever read. And the fact that it is not universally accepted and universally known is a disgrace. Well said. Now, Rich, you also worked with some examples in this, and I seem to remember one of them was where you took like a four decades worth of returns.

And change nothing except the order of those returns. Tell us a little bit about what happened. So this is such a, I totally agree with Dave here, mandatory reading it should be. Because geometric wealth is what we should all be aspiring to. And all of the traditional metrics, many of the traditional metrics we find in finance, they avoid that part dependency. So the popular statistics, you'll hear about sharp, sortino, variance at risk, standard deviation, even what you find in the capital asset pricing model with modern portfolio theory. No matter what order of returns you get, it doesn't change the outcome. You still get the same order in variance statistic. But what you find with part dependency is that the order is essential. The sequence of returns is essential.

And what I found when I looked at 40 years of history, reshuffled it, because wealth production is a multiplicative process. So you're dealing with something multiplied by an accumulated balance at each step along the way. And that accumulated balance really defines wealth over that. What

happens to that accumulated balance? Does it deteriorate significantly? Therefore, the impact of compounding is significantly compromised. Does it accelerate rapidly, which means the impact of compounding is significantly enhanced? What's happening to that wealth? They have accumulated balance. And if things change along that path, Stuff is taken away taxes all of these things stuff is added Redemptions are added into that wealth path. It totally changes the path between the world of what I call additive returns and the world I call path dependent returns So what I found when I reshuffled those returns in the process all of the standard metrics sharp Sortino, variance at risk, standard deviation, they all stayed the same. But what did change was the wealth. We ever had a decline of 30% and it ranged up to a decline of 84% dependent on the trajectory atop.

This isn't a spread we're talking about. This is the difference between a bad year and not being in business. So it is such an important property. And you know, Mandelbrot, of course, was right to talk about the fractal nature of markets and the scale and variant nature of markets. But a very important sequel to that is the geometry of wealth production. finances ignored both of them for many, many years, and they are so important. So, you know, arithmetic returns are the world of additive returns. But what we're talking about, wealth is a multiplicative process. contingent conditional process over the course of time. And the path you take has serious implications regarding the ultimate worth you end up in. If you take arithmetic returns, you know, I've achieved a hundred percent return over 10 years, divide that by the 10 years. That's a 10 percent arithmetic return. The reality is that my path is never like that. I do not get the straight line path.

I get deviations from that path. And those deviations, the volatility, starts creating a different trajectory between the geometric path and the arithmetic path. And as volatility increases, as Dave talks about volatility drag, you find that that difference increases in a compounded manner and so that you might have two very similar arithmetical or additive statistics saying it's the same thing but the path has totally different trajectories based on this this volatility principle that takes a geometric path away from the additive path of the arithmetic mean. Rich, you already mentioned it a little bit, but I'd like to maybe talk a little bit more about it from Dave's perspective. Right, exactly. Sadly over here in Europe at the moment, we have a lot of forest

fires. I know you have them in Australia as well during the summertime. But anyways, you have used forest fires very actively in your...

analogy in it. And it's a very important one because it kind of helps people understand kind of the dangers of things being calm, I think. So can you talk us through it so that we fully embrace that concept? Yeah. So it's based on a physics theorem, a subset of chaos theory known as self-organized criticality as sort of advanced by a wonderful physicist by the name of Parabakh. in one of the, what I would say is probably the most important book anybody will ever read called How Nature Works. And it's about how nature works. It turns out most things in the world are nature. And so the forest fire is the simple analogy that most people are familiar with that, you know, the forest risk, the fire risk grows.

as the forest goes longer and longer without a fire, because you get more clogged brush between the trees. And so as we spoke about earlier, diversification in your portfolio, diversification of your forest solves the problem of a lightning strike that hits only one tree. All your other trees are fine. The risk is when that fire spreads from that tree to the next tree, Because of the connectivity, closeness and clogged brush and tri-brush between them in the financial markets, that would be leverage. And so correlation is the big problem because your diversification lets you down and the fire spreads from tree to tree to tree. In physics talk, what might be known as the percolation threshold. At what point does the density of trees, the density of the forest become dangerous?

And you can think of everything behaves that way. Avalanche is another great example. An avalanche, the longer you go with snow building up, the more likely you are to get an avalanche. Does it mean there's not likely to be an avalanche? It means it's more likely to be an avalanche. And very much to Rich's point about endogenous risk, you don't need an exogenous shock. What causes the avalanche? The last snowflake. One tiny little snowflake. is all you need and that's enough. Now, financial markets very much behave this way because of this reflexive nature and the building, the waxing and waning of imbalances and corrections. And so that's very much how we think about it and talk about what we do when we're out helping people manage the uncertainty of risk. Again, people tend to be quite good

at managing the risk that they're measuring, but the assumptions they're making about the stability of the correlation, the stability of the forest, because they've diversified well, they tend to leave that out. And they say, but Dave, you don't know which trees we own. And I say, I don't need to know. I just need to know where the most dry brush is. And as it turns out conveniently, don't tell anybody this is a little bit of our secret sauce. You know, the biggest fire risk given the nature of how financial markets work. Again, back to rational accounting men and price insensitive players is where the insurance is the cheapest. Because again, things like value at risk are measuring the frequency, a short-term frequency of past fires. And it says, well, there has a bit of fire, so there's no risk. There hasn't been an avalanche. There's no risk. But that's actually generally where the risk is the highest. And so, This is just a great way to look at anything and everything. And the mistake that everybody makes is, oh, it's been calm for so long. It's been calm. Why do you think that's going to happen? Why do you think that, you know, having survived all the way from 30% of debt to GDP to 130% of debt to GDP, and nothing's blown up. So why shouldn't we worry about it?

All the more reason why you should worry about it. Obviously, the imbalance that fragility is building, and that's exactly where strategies like ours and yours is looking for protecting, capturing that tail that the flawed methodology doesn't recognize exists. Yeah. No, no, absolutely. From the world of trend following rich, we think about position sizing and we have four meters for these things. What does this concept, what does that look like when we go into that world? The first thing I need to say is that when people experience a calm market, they think about equilibrium based systems and they think about calmness being sort of almost at an equilibrium state, not much volatility occurring, but that's That's not what markets are. Markets are not closed systems that equilibrate. They're open systems and what I call it the boundaries of chaos that is sitting right out in the frontier and the boundaries of chaos. New participants are always coming in. New capital is coming in, going out. Participants are leaving, coming in. It's an open system. They're far from equilibrium based systems. And when you experience calm in these non-equilibrium based systems, This is not the absence of energy. This is the loose wobbly rubber band being stretched and What that is is is build up of risk that the metrics the

traditional metrics are not picking up It's the undergrowth as they've talking about that's slowly progressively building up to this state of self-criticality and this is what happens with this rubber band stretching so Quiet markets are certainly not where risk is absent. They are being hidden. They're not being expressed by the traditional metrics. But what I've found in my research is they are the deepest coupled negative feedback states in the entire data set. And what I'm meaning there is that in these calm periods, you get markets oscillating. It's not that there's an absence of energy. It's just that their oscillations are cancelling out. Now what happens?

in this state and I'll explain why the risk builds up. Because let's look at some very popular models that are applied to markets. Let's look at volatility, a position-based sizing or volatility targeted. They will have a period of calm and their models mechanistically increase the position size in that period of calm because they're saying it's a safe market. It's not a volatile market. So mechanistically that builds up position size. the energy is going in, in, into the market, but also behaviourary because the markets haven't been expressing their trends or higher volatility, you find that people are behaviourary are starting to leverage up in that environment because they want to create a living in this environment and therefore they're behaviourally they're encouraged to leverage up. Now, of course, we know that, you know, volatility clusters, karma gets calm.

and volatility begets volatility. But what people assume is that the calm is going to, you know, it often continues for a period of time where this mechanistic and behavioural tendency feeds leverage into the system, feeding leverage. So when you do get the shock, it's a small perturbation out of that calm period. Suddenly you get an explosion out of that calm period. And that's where the risk events occur. So what I found in my modeling is that The majority of risk events don't come out of volatile regimes. They come out from calm regimes where they have a volatility shock. And a good example is the Japanese bond market in, I think it was 2018. It had trailing volatility of a fraction. I think it was about 0.9% of trailing volatility. And that authorized 16.4 times leverage.

And therefore, when a move occurred that was only I think a 0.6% move, it took 9% of the capital away. And this is this impact of leverage coming out

of quiet environments. They're the most devastating environments. So do not equate calmness with safety. That's one of the fundamental pitfalls, I think, of so many of the risk metrics that we find that are accepted as rational reasons why we should increase our position size. The high-sharp ratio, well, that's actually authorizing a higher leverage, and that's exactly what you don't want to be doing. We'll get to that later, but this is where finance really needs to carefully look at what they're doing. Let's stay with leverage for a little while, and I don't know necessarily I think I want to leave it as an open question to you, Dave, to maybe talk a little bit about what leverage exposes or the impact dangers that we, you know, when it comes to leverage, you've obviously already given a great example of that already with the star, I forget, was it SLK?

Anyway, to talk a little bit about leverage kind of free flow a little bit on that. Yeah, again, back to what we've discussed. It's loss absorbing capital is sort of in a sense, one of the many stranger tractors that are pulling markets around. And so leverage is obviously reducing the capital that's applied to a potential loss. And so as that leverage builds because of the rules, the regulations, the misunderstanding or the misapplication of Gaussian risk measures, volatility is risk that creates the fragility in the system. And through the active suppression of volatility, central banks, governments, policymakers, they want to take advantage in rational accounting man. They want rational accounting man to lever up that which they want to grow economies, house ownership, various assets, government bonds. And so they suppress the volatility and then mechanistically rational accounting man in sort of his form of non-price sensitive participant goes and applies leverage to that. And he's optimizing to the short-term returns, probabilistic returns, accounting returns of a levered thing that reduces the capital protecting that risk that then leads to the fragility that should the fire start to spread, that's where the problem is. And that's where fat tails come from. And so I say all the time in our world, you want to own and invest in things with thin tails, things that have high natural volatility, where you're getting rewarded with the upside fall, for the downside volatility risk you're taking and then construct your hedges with things that have fat left tails, that have fat tails that have artificially suppressed volatility and attract leverage, leverage explicitly in the form of banking system leverage or repo leverage and implicitly in the form of volatility selling. And those things are the

things that are gonna have surprises and fat tails, fat left tails, some fat right tails as well, of course.

right? That you want to go and position yourself in our case through optionality and convexity and in yours and Rich's case by your signals and your trend thing saying, hey, this trend's pushing towards something that has that signal that that's got a fat tail potential. And the good news is is, you know, Rich, again, you guys talk about all the time, you only need a few fat tails, right? You don't need a whole bunch of winners. You just need a very few winners and then just constantly managing your losses to be small. And, and, you know, that will dwarf all the other activities that you're involved with. You just need, like, you guys were talking about cocoa, I think last time you were on the call. I mean, you know, I suspect for years and years and years, none of you thought cocoa was a big deal in your trend following diversified portfolios. And then all of a sudden it was, was the, what's the big deal? That's exactly sort of the way we look at the world. And I'll argue that's driven by leverage, by imbalances created by inappropriately applied risk methodology, accounting rules, non-skin and the game participants. Now, you mentioned an important word actually, I think, optimal. Because I think a lot of people try to kind of be optimal in many things when it comes to investing. But before I let you lose here, Rich, and maybe you talk a little bit about the kind of the dangers of that word. When I think back, you also mentioned this thing, Dave, about the trend following world that we buy these breakouts or we sell them or whatever it is. And I think a lot of times, I'm going off a tangent here a little bit, but I think a lot of times when people think about trend following, they kind of think that's kind of the secret source that we discovered this idea that you can buy a certain day break out.

I would argue, and I have a feeling that you might agree, but I don't want to be assuming this, but I actually think the whole secret that trend follows brought to the table back in the 70s is that they understood how to manage the risk. And they understood how to size positions differently to what had been understood up until that time. So anyways, don't want to say that because it does tie in to leverage and all of that stuff. But anyways, Rich, what I really meant to ask you is what's the problem of being optimal? Okay, so the problem of being optimal is that it foregoes the room for being

wrong. So let's look at the optimal bet. Let's look at the Kelly criterion. So the Kelly criterion...

It's looking for perfect information. It's asking a specific question. Given an edge, what fraction of your capital should you state to grow it as fast as possible? Now, when you look at it, when you're looking at the optimum condition, it's putting you at a peak. Because if you go over that peak, you become suboptimal. And what you find with a volatility drag, the going over that peak actually puts you into risk of ruin very quickly. If you under bet and you don't go quite to that that the Kelly limits That's taking significant risk of off the table. It's giving you room for error giving you room for not being correct Now this is the problem with the Kelly criteria. It's it's assuming perfect information and it's one another sort of addition to this Gaussian world And it's a false premise because what you find is that markets are non-stationary. You find that there's significant noise in the market. Now, what people don't realize is that when they are choosing the optimal bet, they are applying an estimate. Not the reality is an estimate. Given their sample size, what they're looking at, that's what they're placing as their optimal bet, their estimate.

As you know, when you start introducing fat tails into estimates, your sample size needed to make that a fair and reasonable sample size explodes out. Now, the Gaussian world, it reduces it to a small sample size needed to be a degree of certainty. But when tails are included into that sample, it explodes the sample size out. So you never have enough sample to be anything other than an estimate and there is always room for error in that estimate. And so because you're trading at that peak, there is no room for error because that's what you give up being optimal. To be optimal means you've got to give up that room for error. That's the biggest mistake. It's putting you at the knife edge.

It's it's increasing leverage to the point where and I've done this myself. I've I've looked at Kelly base position sizing models with our trend following approaches Niels and Dave and I've compared it to very simple position sizing rules that are regarded as you know equal weighted risk betting fractional position sizing that we adopt in trend following world and it outperforms like the. the Kelly approach when you start introducing things

such as regime changes, sufficient noise to your estimate, because the noise is often bigger than the signal that they're looking for to develop that optimal bet. And this is the problem. long-term capital management phase. So they traded that they used optimal trading positions for as long as the assumptions held and are very successful over the time the assumptions held. But when they went over that limit, they went quickly to risk of ruin. That's going over that peak I was talking about. That's the problem with optimal. To be optimal, you've got to have perfect information and we never have that perfect information in these complex adaptive systems.

You know, there is in podcasting something called the optimal duo for having on a podcast show and I think we're achieving that today, but there we are. That's completely different concept. No Kelly here. Anyways, we're going to move on to something where I think over the years people have heard kind of analogies about football and should a trend follow. Is he a striker? Is he a defender? Is he a midfielder? But Dave and you and one of your colleagues, I think Nicholas, have kind of taken this football analogy and obviously with the recent World Cup to a brand new level. And I know you published this this morning. First of all, everybody should follow both Rich and Dave on LinkedIn.

Twitter, wherever it is. But this kind of takes it to, I think, a new level. I've only skimmed it. So I kind of need also for selfish reasons for you to take us through this new World Cup scoring system, I guess we could call it. Unfortunately, that was Nick who just left. We wanted to get rid of him. Yeah, I saw he leaving there. Yeah. Yeah. So forever we've used a football pitch picture. in a lot of our presentations, if I speak publicly almost always because I want to frame what I'm talking about and get it in pictures heads. Because the world, the financial world is so, you know, emblazoned in their mind the Gaussian distribution, right? So we draw a football pitch and we draw the Gaussian distribution over it and explain that that's a measure of frequency. That's a measure of where the ball spins time.

on average, the balls at the center of the pitch, like the average return of monthly S&P. It's the average. It means nothing. The average is meaningless. It is a wasted statistic. And then we draw what we call an entropy curve, a Shannon's entropy curve, a measure of information value.

And then we shade underneath that the long-term contribution to compounded returns of the S&P and show that the two percentiles of returns, the 10 worst months out of 480 months and the 10 best months out of 480 months contribute almost all of the returns. And just like in a football match, it's what happens in the penalty boxes that batters, not what happens in the center of the pitch, right? It's goal prevention and goal scoring. And so we've been giving that presentation forever. It's certainly longer than you guys have known me. I've been giving that presentation. And so Nick, who is you know, who takes geek to a different level here, uh, said, I'm going to build a model and we're going to track it through the World Cup and see if this story you've been telling forever Dave is true. That what happens in the penalty boxes is what drives the outcome. And we did it. And, you know, thankfully to nowadays FIFA provides unbelievable data and there's all kinds of improved data with the chip that's in the football doubt. All right. And, and we came up with what's we've called the convex goalkeeper theorem, and without going into all the mathematical complexities, the gist of it is, is it comes down to three layers. This is how people win a non-urgotic football tournament. Now it's non-urgotic because if you lose, you're out, right? You hit the ruin barrier, as Rich would say, the insolvency barrier, you're out. You've got to stay in to keep going, right? So it's a very non-urgotic path.

Layer one is defensive resilience, right? Are you becoming more defensive as the ball moves from the midfield to your penalty box? And we measure it by the 18 yard box and by the six yard box. Now, obviously the ultimate defensiveness at the very end are saved by the goalkeeper, but we have other measures and metrics in there about how we're measuring sort of a positive convexity of defensive resilience. Likewise, the other part of the game is scoring goals. Are you becoming more effective, efficient in your attacking as you approach the other 18-yard box and six-yard box and goal? And it turns out, layer one, most important, layer two, next most important. But then we found a layer three. And layer three was individual tail risk, if you will.

an individual performance above and beyond what could be recognized from simply the dominance in the respective penalty boxes. Messy making nine crosses in the last two minutes, right? Or Bellingham's last second two

goals, et cetera, right? Think of that as a tail risk that depending on which side of the game you're on is either a negative tail or a positive tail, right? And so staying in the game, such that that tail can be beneficial to you, very important, right? That can make a difference in the match. And so it's a fantastic, the model is really fantastic. We put the paper up, a shortened paper. There's a full paper that goes through every single game over the whole course of the thing. We've put a shortened one up with a few key examples. And so one of the best examples, and you guys will love this, is the England-Argentina match, because you can divide that match into two matches.

It was a very normal batch until England went up one nil and then England did, as they say in England, parked the bus. They did what is so common in the investment industry. They thought they could reduce risk by capping the upside, by only playing defense. They did what in option space we might call a covered call strategy. Buy my bonds. They sold call options. They bought more bonds, right? They did 60-40. They reduced speed. They drove slower. And yet they're still exposed to the tail, right? 60-40 can still lose money. A covered call strategy has still got all of the downside. All you've done is pour gold. The ultimate objective of good defense is attacking, right? You can't separate, you know, a kid to rich is such a beautiful story. You can't separate the parts.

You know, the hole is greater than the sum of the parts. The parts must function together. You must be convex to both sides of the magnitude where it matters. It's not the frequency. It's not what's happening in the middle of the pitch where the ball spins most of the time that matters. And I found it wonderful watching. And obviously, unlike most Americans, I've played soccer as we call it in America my whole life and coached soccer when I was a younger man, some interesting teams. But even watching this World Cup think how many goals and ritual love this? How many goals are the result of a mad scramble? So many right very few are actually a smooth perfectly orchestrated movement so many are a rebound a ricochet a Stumble a fall right and the the act of getting or preventing them is about this unknowable emergent property of defensive resilience and offensive effectiveness. And if you're just trying to play by the numbers, you're not gonna find that. And so it's a really interesting model. I'm gonna try to, in

fact, I told some people today, I'm gonna speak at a CBOE event in Hong Kong later this year. I'm gonna try to somehow form it into one of my presentations, tie it together with total portfolio approach and stuff.

It's really fun. Everybody should go look at it. Absolutely. I completely agree. Absolutely. Now, you've already alluded to, Rich, that we are skeptical of SHARP. Obviously, Dave has written tons about the SHARP world that we live in. So, you know, if you have two portfolios with the same SHARP, what do we not know from looking at those numbers? Look, before I get into that, I've just got to say, Dave, I can see a sequel to Moneyball coming up. We're applying convexity now in the game. So looking forward to you doing that, Coach, and seeing how that goes. We talk about that all the time, actually, Rich. Let's spot on. Look, I also want to sort of add to Dave's point. And Nils, you'll be interested in this because let's talk about trend following.

When I ran my models over 40 years, you find that October 2000, the date, that was a fantastic period for trend followers. It was the single largest month in the return for the entire history of 40 years that I looked at. It accounted for 3.4% of total portfolio wealth that one month, October 2008, 3.4%. But you know what happened to the SHARP ratio over that same period? It went from 0.914 to 0.931. So it paid two cents in the dollar on you. So the SHARP penalizes trend followers. They just wanted to say that. There's no fault of the ratio. It's not deliberately picking us out. But because when we're employing convexity, the SHARP is perhaps the worst measure to use to assess that. So SHARP.

to me fails in two ways. The first thing it does is it doesn't let you know the system's failure modes. Now I want to give you a good sort of example to consider. So consider two portfolios both with a sharp ratio of 1.5. So the first is a single concentrated position. Leave it up using Kelly resting on one estimate of one edge. That's one of the portfolios producing a sharp of 1.5. The second is a wide book of small positions that are only loosely related. No individual position is carrying the load, unlike that single leverage estimate. That also has a sharp of 1.5, same number. So the metric sharp itself calls them equal. because by the time it's calculated, the entire construction of SHARP has been compressed into two figures, an average

return at the numerator and a volatility in the denominator. And every trace of how the portfolio was assembled has been discarded, apart from those two values. So the number records where you arrive, but it doesn't record how you got there. And it's the second thing that decides whether you're still standing when an assumption breaks.

It fails, sharp fails particularly in two ways. It tells you effectively the load rating and so consider a single strand over a canyon. We've got a single strand rope that's holding a particular load. That measures a sharp of 1.5 compared to a thousand strands or individual strands or loosely assembled that has the same sharp over that canyon. Now, we've got one failure mode for the single cable, but if we get a single failure mode in the ensemble of loosely correlated estimates, we only get a failure in one and 999 is still keeping them up. So this is where the SHARP doesn't reveal this property. And it's only through understanding the construction, how things are, you know, the architecture of the construction of the portfolio.

That's what's telling you what are the foundations of that architecture? What are its dependencies? Is there one failure mode or is there a potential for many failure modes in your portfolio when crashing down? You don't get that out of that sharp estimate. You get that from a knowledge of the construction. And that's something none of these metrics is going to give to you. This is why I believe Dave and I are what I call engineers. We are architecting convexity. We are architecting this geometrical progression that mitigates our downside risk and exploits opportunity. This is this break, the break system that Dave talks about in his Formula One. It's mitigating the volatility drag and getting the beneficial opportunity, exploiting the opportunity because you've got good breaks. That's exactly this principle. That's all I've got to add to that. Sure, sure.

Now, we've talked a lot about, you know, risk and how to manage, how to mitigate that. And I think, I can't say that for sure, but I think a very popular way of thinking about mitigating risk is just diversification. So my question to you, Dave, is, does diversification really protect you? Yes. Sometimes. Right? Just not when it matters the most, right? So again. Traditional diversification is making assumptions about correlation, making assumptions about ignoring the connectivity between the trees. I wrote one

of my papers once years ago pointing out that diversification across a whole bunch of things that any one of them can cost you all of your capital is probably not a good idea, right? Because that any one position can wipe you out. If any of them can go to infinity, infinity loss will wipe you out.

On the other side in what we're doing, right? The more diverse vacation, the better, because everything that we do as best as possible, we want infinity as a potential benefit. So I only need to get to Rich's point, your point, you only need one month to make an enormous contribution, right? You only need one trade. If Coco goes to infinity and you're on it, you've made your budget forever. And so the same thing with us, a big part of what we do and how we construct the convexity for our investors as much as we possibly can. We always want infinity on the other side. I don't know if you're listeners or you guys are familiar with the St. Petersburg Paradox, which is a coin toss game that has an expected return of infinity, but a very low probability of achieving it. But if I can at the right price get someone to commit to let me play the game, for an unlimited number of hands. And I've always got infinity on the other side. Well, eventually I got a pretty good chance of him's begging me to unwind the commitment that I can play forever and ever and ever. And that's a big part of how we think about things. So diversification needs to be understood for what it is. There are beneficial diversifications. There are faux benefits of diversification. And there are dangers of diversification.

where you don't have bounded understood downside across the range of diversified and possibly reflexive items. Yeah. Rich, you talk about this using words, the fact that diversification is kind of more the engine that what shields you talk us through that idea. So Niels, what I found in my research? is that diversification for us classic trend followers is it's giving us exposure to more right-tail opportunities. To me, using diversification to mitigate risk, as Dave says, can be good, it can be bad. Here's an example. We've got those thousand strands going across that canyon, looks diversified, but imagine if they're all connected to a single anchor at each end of that canyon. There is still one failure mode there, but it looks diversified. And this is a problem, you know, you might borrow short and lend long across a dozen markets during calm regimes and think it's

fantastically diversified, but you're exposed to a single interest rate bet effectively through that.

I think correlations, what I've found in the research, and this is really interesting to me, it's research I've done. I haven't really promoted it. But what I've found is that most people think that markets during calm regimes, ordinary regimes, you know, tend to move independently of each other. And then during periods of stress, suddenly they become correlated. Well, I'd like to differ in that opinion because I think that markets are always permanently coupled. Now imagine this. So you know how I was talking about ecosystems of participants. Now the individual agents in that system, they are really holding a single position in anything. They are diversified across products. Now that model that they are applying diversified across products is creating this permanent coupling. So as long as diversification exists in the market and most investors apply it, I think we will find that the markets are permanently coupled. But what does change is that the phase of where it's sitting in that coupling. And what you find during calm regimes, we get this oscillation effect, cancelling effect. This is what creates this apparent benefit of diversification. Diversification works during calm environments, but it works specifically at the wrong time. You want it to work during periods of crisis. But when you get into periods of crisis, the coupling goes from oscillating to trend.

and you suddenly find that everything becomes correlated. This is this permanent coupling through diversification, through agent-based models, all sort of spreading out across asset classes. A commodity trader is spread out across his commodities. A global macro trader is spread out across currencies, bonds, all of these things. These forces, this coupling, is what there's this coupling of extremes and where the correlation is. Correlation is an expression of where the coupling sits. It's always on. but people assume it's off during good times. So the problem I face is that you've got to use architecture here. So go back to that example, a thousand strands, one anchor, we see the problems of that. But if we've got a thousand strands connected to a thousand separate anchors, we've got a totally different proposition there. So when we get, you know, we're looking for Not necessarily correlation properties. We've got to know the architecture. We've got to know what are the dependencies that exist in our portfolio.

You're not going to get that from the statistics. You're not going to get that from correlation properties. You've got to understand how the architecture, how it's created, what are the dependencies in that architecture. Another really interesting point.

is in my research I found that diversification doesn't thin the tails. So if you imagine everyone's assumption is that with increased diversification the impact of tails dilutes. But what I found is that with increased diversification we get an initial drop in kurtosis and then a ramping up of kurtosis because what you find is that the tail properties because they're fractal in nature as you get bigger and bigger samples, you get exposed to more large things. This is unlike the proposition that with increased diversification, the large things dilute. No, not in fractal structures. If I get a big tree and I put a sample over that tree, a small sample, more than likely I'm going to get leaves and small branches, nothing material. Increase that sample, I get more exposure to the trunks.

to the large branches, increase it further, I get the entire tree, I get the major dominant fractal structures of that entire tree. This is what happens with diversification. So what I've found in my studies, as you increase the diversification, you get more exposure to the right tail. I'm not using diversification here as a solve all for the correlation argument. I'm not saying it's going to... significantly reduce your volatility. But I am saying that you get more exposure to the fact fractal properties of the market. And because of this, you get more exposure to the right tail. And an example, who would have been sitting in cocoa? Most traders don't even consider cocoa. But because we're maximally diversified across all of these, we had a position in cocoa.

That was the thing that exploded. Now, those people with a small subset, small portfolio, which would have had the majors in it, the currencies, the bonds, that sort of stuff, they had no exposure to cocoa. We had that idiosyncratic exposure to cocoa because we are maximally diversified. And this is what I find, this leptocurtic property of markets, the fractal nature of the markets means that as you increase your sample size, you find that the large things actually get larger. And that's why your drawdown is always bigger ahead of you, as will the greater returns will always be ahead of you

as you increase the sample size. And I've taken 125 years of market data from the Dow Jones, combining that with the S&P 500. And when I look at the major material moves there, you find that the The major material moves, the tail properties that Dave is convincing us about, and I totally agree with him, they are the major drivers. That's where all the information is. The rest of the movement is just noise. It's perturbations. That's all it is. These drivers, as you extend the sample, you get access to the drivers of the system in this fractal structure. Fractal is a scaling variant. You can go down in the structure, always stays there.

You can go up and the structure always stays here. But as you go up, you get access to bigger, bigger, bigger, bigger structure. This is what happens with sample increase in diversification, at least in my research. Sure. Super interesting. Now, we've got a couple of questions left that I'd love to talk to you about. Next one, pretty important, I think, because what people will have hopefully gathered from our conversation so far is that asymmetry is important in the way we view opportunities, markets, et cetera, et cetera. But you guys come at it in a different way. Dave, you buy this asymmetry. Of course, Rich and other trend followers, we built it into what we do. Let's start with you, Dave. Why do you prefer this implementation methodology?

Well, a couple sort of two things that I would say differentiates how we go about it. One, I'm creating the trend following type benefit for our investors as opposed to running it ourselves. So our investors, we are freeing up capital that they have traditionally tied up in poor diversifiers. by creating explicit risk mitigation breaks so that they can go out and put that capital to work, participating in the positive drift or long-term trend of asset price inflation in compounding assets. So the positively correlated side were allowing them to carry more of on their own books. Well, we focus only on the negatively correlated convexity asymmetry that we need to constantly work to maintain sensitivity and attachment as their assets grow in in in size compounding and price moves through time through this drift and Appreciation asset appreciation through time. So we need to then actively manage the negative correlation. So one, we're trying to create very much what you guys are doing, but the combination of what they do for themselves and what we then provide them creates the long gamma, if you will, on their combined portfolio, as opposed to us running both the positive

and negative correlation in our books, as you guys would as more of a standalone strategy. So that's one, two. Now, this one to me is the big, this is my big thing, because it's what I do, all of that asymmetry, all of those non-Gaussian distributions that you guys capture, the movement of the actual underlying realization of, we're in the business of buying that implicit leverage that allows us to overlay that on the mispricing. So we're buying long dated, term periods of the mispricing of the underlying distribution, which itself is driven by the supply of volatility dominated by non-price sensitive participants, predominantly structured product vol supply, that is itself mispriced relative to its own distribution. And so you have this sort of double whammy, how are you triple whammy of a price for the flawed distribution of the underlying over a long-term period of playing for that, that itself is mispriced, that is constructed in product construction that has inherent in it an added level of asymmetry, right? The non-linearity of explicit option pricing or volatility type products. And so you get sort of this triple whammy.

And then again, to our, my St. Petersburg paradox example, I'm not just getting to play the game one time. I'm committing to play it for three years at the same price every time. And so not only might I get to win the large tail one time, but that one time almost certainly raises to the realization of somebody who's provided this non-recourse leverage that they've mispriced it. So they may not want me to get to play the game at that price again and again and again every day with infinity on the other side for the next two, three, five, 10 years. And so that's how we're going about it in hopes that we create the recognition of the goalkeeping efficiency or the breaks that allows then our investors to go out and be much more aggressive. And so exactly the conversation we were just having about diversification. And this is, again, another dredgism that goes back to my bankers trust days running bankers trust businesses during the Asian crisis out here, which was a pretty wild time, more wild than anything I've ever seen. And I've seen almost everything that was wild that when you're positively convex, more risk is less risk. Exactly what Rich was saying. When we're positively convex, we want the whole tree, not just a branch.

And we're happy to have a room full, just exactly like you guys are, of opposing views. We don't have actually any views. If somebody can create

great convexity that is positively directional and somebody else wants to create convexity that is negatively directional, that is bearish or bullish, the more the merrier, right? You know, we're reducing risk and capturing more tales all the better and go for it. So it comes back down to that, obviously, and you know, this is what Jim and I talk about all the time when we're together, is price. What's the price? That price again is heavily, heavily driven by non-price sensitive participants and flaws, if you will, or I don't know if you want to call them flaws, they just are what they are in the application of regulatory risk and accounting rules on the main market participants that allow them to recognize enhanced return, these repackaged yields of captured option premium, and not necessarily capitalize the tails. And they don't have skin in the game. It's not their money. And so they're not thinking about the possibility of what could happen. They're measuring the regulatory probability of it. They're happy to, another example I use all the time, Rich, is we're going to a casino and because the the croupier is getting 10 cents of every dollar he makes. He's trying to attract people to his roulette table by offering 101 payout. But what I'm trying to get him to do is commit to that 101 payout for a thousand spins, not just one spin at a time. And when he figures out that he's got a problem, he's likely to try to buy me out of my original agreement. And that's where the big asymmetry comes from, volatility or volatility, convexity. Yeah, that's a great way of explaining it.

share some thoughts on this, Rich, and why Trinfollows do it slightly differently? We do it slightly differently, but I think Dave and myself particularly, we're playing the same game. So we're engineers, we're trying to architect the optimal geometry of returns to survive and to capitalise and exploit opportunities. So If you imagine, when you look at the market structure, the markets themselves had very little skew in them, but through the processes of both Dave and a trend follower does is that they truncate that left tail with their return distribution. That's the brakes. They truncate that left tail as we do it with our stops. We do it with our small bets. Dave does it with a small premium you pay for the option. We're truncating that left tail, but we're leaving that right tail open, the positive skew, we're leaving an unbounded opportunity because no one can predict what these markers can do, but the fractal structure...

tells you that there are things that we've never experienced before which we're going to see and they can be exploited. This is the accelerator on the racetrack. If we protect our downside, if we protect that volatility drag of the downside and we exploit that opportunity, Dave and our solution, his options and our trend following, it's doing the same thing. If you look at a long position with a stop, that's what we adopt, that's a synthetic core. A short position was a stop is a synthetic put. You get the shape of an option in trend following, but we're just not paying an explicit premium for it. But there is a premium. We are paying for it. It's a loss as we incur. It's a foregoing growth that we lose. There is a premium that a trend follower pays. They often say, oh, but we don't pay the premium, like the long vol specialists out there paying for their options. We do pay it.

We do pay it and we're paying it through the stops that get triggered out to forgo future opportunities. We do get those and it's a psychological cost. So Dave gets a line item. you know, where he's got the premium to pay for his option. That's his line item. We don't necessarily have that. We can't see that line item, but it's there, trust me. It's there in how we experience things. We experience lots of losses, whipsaws along the way. That's the price we're paying. That's our premium we're paying to keep that truncated left tail and to exploit the right tail opportunity. So I see that both of our approaches is architecting a profile to optimize geometry because path dependence is so critically important. That's what we've expressed over this podcast. So, you know, I can only see similarities here. You know, I might see a slight difference in that. I think Dave trades what I call disk.

continuity. In other words, when we put a stop in, there's no guarantee that that stop is going to be triggered. We might get a gap through that stop. We'll still have small bets with maximum diversification. Dave is guaranteed with the explicit premium. So he immediately gets that guarantee. We get it through what I call sort of duration over the course of time, very similar, but slightly different approaches. So yeah, I can only see similarities. In fact, what I see is a benefit of perhaps combining both approaches together to get the both worlds. But that's how I see it, Nils. No, absolutely. We advocate that to everybody we work with, right? Is that there's path dependency and there's, again, any way you can add positive convexity, add positive convexity. And there's a whole bunch of ways to go

about it. And the fact is that the opportunity to capture cocoa, we didn't have cocoa because nobody came to us with a volatility structure and volatility supply that suits how we do it.

Now, we might have things that nobody ever thought anybody would do, but that's because there was a volatility supply. But we're limited in what we can do because there needs to be efficient volatility markets for us to do it in. And there needs to be efficient volatility products even above and beyond the simplicity of what might be exchange traded and stuff. So they're very complimentary. They're different parts of the distribution. There are different types of scenarios, the breadth of what you guys can do. We can't match in terms of markets. The attachment points are going to be different. Every way you can add convexity, add convexity. That's the answer to everyone's problem. Yeah. I mean, there's a lot of similarities. I was smiling when I read your June paper where you have this chart where you show what happens when you add 50-50 long vol and the S&P.

And of course, you don't just get an average, you kind of almost get the sum. And I use exactly the same chart for trend following. If you add the S&P and 50% on capital, it's not just the average. It's almost the sum of the two. It's incredible. Now, this conversation will air on Saturday, so a day and a half from now. What I'm hoping for is that there are a lot of CEO or CIOs out there who's going to say, whoa, this is interesting stuff. So I guess my final question, unless I think of more, what do you want them to do differently Monday morning? Dave? Oh, it's the same thing. I will always say add convexity, right? So the first thing that needs to change in the investment, the fiduciary investment universe.

is a change in the incentive structure. So we have to somehow shift the incentive to being this geometric compounding path. That just doesn't exist, right? Everybody's getting paid on calendar year, arithmetic returns, time ignoring ergodic processes. So you need to change the incentive structure and get people to understand that the objective is the growth of wealth. Expected return is a nonsensical number and it's destroyed more wealth for capital owners than maybe any other simple thing. Now arguably they have transferred wealth to fiduciaries and so there may be some reason why they stick with it, but that's just me being cynical. So you need to add convexity.

How do you add convexity? Rich and I have talked about optimizing isn't the right answer. It's a learn by doing because you're now building resilience to the divergences from the expectation as opposed to optimizing to a backward looking expectation.

So you can't optimize to the unknown future path. All you can do is put brakes on your car and go out and learn how to drive, go out and explore how to go faster, aerodynamic, stronger engine, different tires, different steering wheel. But before you start exploring that, you better have brakes, right? So you need to start freeing capital. This is sort of my current theme and presentations I give. You need to start freeing capital from the ballast. that you've been using to try to just slow you down. And you want to then use that capital to invest in breaks or free the capital by getting breaks and then use it to go out and find ways to participate in more positive market environments, positive upside participating type assets. And so learning by doing trial and error.

is how you learn how to drive with brakes as opposed to trying to estimate the average lap speed, start trying to extend separation over multiple laps. And it's a learn by doing. Yeah, I've got a follow up question on that in a second, Dave, but Rich, what do you want people to do Monday morning? Well, I want them to pay less attention to those damn statistics and more get into the underlying. you know, foundational assumptions of the system you're assessing. Look at the dependencies, but specific things I'd like them to do. Here's one, cap leverage in absolute terms. Now, what I'm doing here, there's no statistical reason for this apart from what we're doing is we're breaking the automatic link between a quiet market and a large position. So if we cap leverage in absolute terms, we break that link. That's an important one because most of the capital destruction, he says, coming out of quiet markets of too high a position size, too much leverage. Leverage is the destruction of compounding. This is what I found. The second one is size every market to the same risk rather than tilting toward whichever sort of edge looks strongest. Now, this comes back to the selection argument. There's selection bias in our selection of things that have recently performed, et cetera.

by selecting or increasing our position size to a particular asset because of its recent performance, what we're doing is we're striving towards that single cable between the canyon again. We're losing the breadth that we had through the all of the different cables tied to separate anchors. And we don't find in the noise that exists in that market, we're always estimating the return, but it's not enough to give us the optimal bet as we talked about with Kelly. Size every market with the same risk rather than tilting towards whichever edge looks currently strongest. That way you miss cocoa, that way you miss all of these explosions that we talk about. It's not a refusal to think. It's actually the opposite. You still choose your markets. You still choose your horizons and your rules. That's the whole craft of what we do as trend followers. But what you decline is a step afterwards, ranking those markets by apparent strength and betting the difference. So the difference between edges are smaller than the noise involved in measuring them. So a ranking is a signal too faint to trust in my sort of understanding of how these markets work and tilting towards the leaders quietly rebuilds that single cable that we've got to worry about. And the third one is grow by widening the book, not by scaling the bets. So we're avoiding leverage here. So when you want more return, the instinct is to increase size on what you already hold. Now, what I'm saying is I'm not advocating that. I'm saying add markets, add different systems, different diversification.

expand your diversification, don't increase concentration in your single bets. That way you're reducing this reliance on this estimate once again. So it's the difference between a heavier load on one cable and rather than look for more cables to that bridge, more sort of unrelated things that won't bring your portfolio down. That gives you more exposure to the tails of the fractal markets that we live in. And the last one is Enter small, so always enter small and let the market build the position out of open profit. This is where I say, let the market do the heavy work in lifting your performance. Don't use size or positioning or leverage to try and push your will upon the market. Let the market itself do the lifting power. Keep your bet sizes small on any individual estimate that you make.

While you've been talking, I've thought of a question. And when you do things sort of on the fly, it may not come out as clearly as you would like. But let me try anyways. So back in the 90s, we started talking about in this

industry about portable alpha. Now, that really never came too much. But I certainly clearly remember this and the concept of saying, Having an equity portfolio new on top of it would add trend following. I don't think I talked about long vault back then. It probably didn't really exist in the area that I was working in back then. But of course, more recently, it's coming back with a vengeance. We see a number of managers launching these portable alpha return stack products, etc. So this is what I'd like to.

for you to help investors understand or maybe deal with. It's the challenge where on one side, as Dave says, you need to free up capital. Now you can do that in a sense by doing a portable alpha product because you can replicate the index and you can add trend following or you can add long vol and you still have money freed but that line item is essentially it's more efficient. But if you look at it from a traditional perspective, you would say, well, I've got 100% equities and now I'm adding another 100% trend or another 100% vol. That to me is a leveraged product. So on one hand, we say, don't go down the route of just adding too much leverage. At the same time, we're saying this is so much more efficient. And when you combine these two is specifically long vol and trend following.

with your underlying traditional market, you actually get a much, much better product. So how do investors who may not be spending all their time like we do in this world, how do they square the risk of too much leverage with the benefit that we're arguing saying, well, actually in this case, you should do it. If you know what I mean. Exactly. And it comes back to concavity and convexity. Right. So most people are levering. concave risks. So they're levering risks where their correlation is low to good markets and rising to bad markets. So they've got volatility drag in that dynamic, right? The volatility has a negative impact on their geometric path and adding leverage to that will create disasters, the disasters that we talk about in terms of what drives markets.

If you're positively convex, if you have explicitly negatively correlating risk, that is growing so your overall portfolios increasing correlation to positive markets and reducing correlation to bad markets, right? And you think of it as non-recourse leverage where you've gone out and bought an option that allows you to participate in the upside and give the downside to

somebody else. Well, that is risk-reducing as you grow that position. So back to what I said before, when you're positively convex, more risk is less risk. And so you can see when, you know, we play around with examples, well, the more long vol, the more hedge we add, the less risk you have. So you can go take more risk. And then guess what? You could add more negatively-correling, highly asymmetric protection and go take more risk.

And so you can keep going and going and going with this ad infinitum. But adding leverage, recourse leverage, to concave positions is exactly what creates the volatility drag and destroys compounding paths and creates the tail risks, literally. Yeah. Yeah. Do you want to add something to that? I can't really add to Dave, apart from the fact that Dave works in this world all the time, recommending long vol componentry to add to traditional portfolios, fortunately. in my trend following world, I'm 100% trend, so I never have to face this demon that we're talking about. But if it improves concavity, that's a good move. That's how I'd see it. But really what we're doing here is we're applying a Band-Aid fix to a bad return string with that traditional portfolio. It's basically like a Band-Aid.

approach to fix it. I'd say ideally, you know, resurrect the architect and start from Brown principles and construct that portfolio without having to have that demon to deal with. The poor old Dave has to deal with with long vol integration into traditional portfolios all the time. I also remember from my early days in the early 90s in this space that one of my colleagues was using this slogan saying, And of course, again, this was about trend following. I'm sure it could be said about long ball is that trend following allows you to own equities. And I'm just thinking all these years later, I mean, it's probably as true as it's, you know, ever has been yet most investors have not yet embraced any of the two strategies that they need the most. But hopefully.

the last two hours with you thank you and thank you for for for going along here so to speak will change or challenge some of people's assumptions about their investments and so very excited about that and super grateful by the way for the knowledge that you guys have shared in the last couple of hours and I'm sure all our listeners will be as well and maybe a little bit of What I like to do, actually, I like to do something a little bit different here.

First of all, I like to ask a small favor of the listeners, and that is to go and show the appreciation to both Dave and Rich by heading over to a favorite podcast platform, leave a rating and review, because it really does help with more people seeing this. But I like to up the stakes for the first time here.

Since if people watch us on YouTube, they will see that I'm the one who forgot the top traders unblocked merge here. So what I'd like to do, I'd like to make a little challenge here and say, there is a top traders unblocked vest at stake here for the best review that is posted in the next month. So we give it to the end of August so the people have a chance to go out and leave. an amazing review for for rich and dave however since i don't have infinite time to monitor all these platforms i need to ask a favor i need you to email the usual one in for top traders on block dot com i'm probably gonna hopefully i'm gonna regret this with a large number of emails to email you what the review was. Where you posted it when you posted it so that one of my uh one of my team members.

We'll go and verify and then we will either buy democratic vote on the top traders on block team, select the best and get in touch to the email that you send us so that you can get your merge. Or we may ask if we can figure out a way to do it. We may upload them to chat GPT and ask. Chad GBT, who do you think should get this vest? So, upping the stakes for the first time here. Hopefully that will get more people going on that. Anyways, as you can tell, I'm super excited about this conversation. And I really do hope that lots of people will tune in and listen to it. I mean, there are no better people than the two of you to explain these things. So, exceptionally grateful for that. Next week.

I will be joined by Harry Moore from a man HL. That will also be an interesting conversation, not least because I think we may touch also on portable alpha as I have a feeling he's coming out with a paper soon. So we can put all of these conversations into perspective. But in any event, I'm going to give you the chance, as I mentioned before, go and follow Dave, go and follow Rich. They put out some amazing information. But also I want to leave a room for both of you to any final words, anything you... I wouldn't say we left out. I think we've been pretty thorough. But anyways, if there is something that I forgot to bring up, you can certainly add it. I'll

throw one last dredge's about there. And so this is something that amongst my various other activities, I'm involved in a lot of risk councils and risk conferences and stuff.

people often are waiting for me to give them some magic formula of what is risk and what is risk management and what's the right way to do it. And I end all of those presentations with this slide. Risk is about accountability and that there's no exactly what we're talking about here. There's no magic modeling of risk. There's no magic formula of risk. There's no magic regulatory model rule. The only way that risk can be appropriately managed is through accountability, is through tacit knowledge, is through recognition that if I crash, I'm out of the race. And until somehow that gets instilled into all of these non-price sensitive rational accounting men, the system will forever be fragile. And I say this to regulators every time I talk to them since You know, not even I wrote notes saying do not embed value at risk in BIS regulatory capital guidelines and remove accountability from the guys actually making the decisions or you'll get the GFC and nobody's listened to me yet. GFC or not. Well, I also like the fact that you often remind everybody that actually risk is everything you didn't think of, which in itself is a problem. It's not in your back test.

But it's not in your practice. That's right. Anything on your side, Rich, you want to leave with? It's just been an absolute pleasure for the famous Mr. Dredge, you know, big fan, as I've already mentioned. But it was great chatting, great sharing ideas. And this is a great venue for it, Niels. I thank you very much for this opportunity. Sure, absolutely. I'm sure we will do it again. Anyways, to all of you out here, thank you so much for listening. from Dave, Rich and me. We look forward to being back with you next week here from Top Traders Unplugged. And until next time, as usual, take care of yourself and take care of each other. Thanks for listening to Top Traders Unplugged. If you feel you learned something of value from today's episode, The best way to stay updated is to go on over to your favorite podcast platform and follow the show so that you'll be sure to get all the new episodes as they're released. We have some amazing guests lined up for you and to ensure our show continues to grow, please leave us an honest rating and review. It only takes a minute and it's the best way to show us you love the podcast. We'll see you next time on Top Traders Unplugged.

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